Dominik Lentrodt

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Experience

2025- **R&D Laser Physicist at Toptica Projects GmbH**
First and main theorist for the company's laser development. Responsible for product performance predictions, numerical simulations, data analysis, and versatile R&D problem solving via modeling and physical process identification.

2025- **Founder at revviews UG (haftungsbeschränkt)**
Software company aiming at providing an online marketplace for scientific peer review. Currently under development at revviews.org. Experience with AI coding tools, web-development, and online payment services (Stripe, Paddle).

2021-2025 **Postdoctoral Researcher, Georg H. Endress Fellow**, Freiburg University & Basel University
Independent fellowship funded by the Georg H. Endress Foundation.

Education

2017-2021 **PhD in Physics (Dr. rer. nat.)**, Max Planck Institute for Nuclear Physics & Heidelberg University
Title of thesis: Ab initio approaches to x-ray cavity QED.
Topics: Quantum physics, X-ray physics, Optics
Grade: Summa Cum Laude

2012-2016 **Master of Science & Bachelor of Arts**, Gonville & Caius College, University of Cambridge

2009-2012 **German Abitur**, Maria-Theresia-Gymnasium, Munich

2010-2011 **"Frühstudium Informatik"**, Technical University of Munich (TUM)
Undergraduate course program for excellent school students.

Skills

Leadership

Project leadership
Example: Conceived, planned & coordinated multi-national university collaboration (Freiburg, Lund in Sweden, Erlangen, Hamburg, Jena) to perform an experiment at a large-scale science facility (synchrotron source PETRA III at DESY in Hamburg).

Student supervision
Scientific supervision and co-supervision of PhD, Master, and Bachelor students.

Presentation & Communication

Scientific presentations
More than 30 presentations at international scientific conferences, including 11 invited talks. 10 poster presentations.

Collaboration & Negotiation
Example: Participated in setting up and negotiating multi-university collaboration for a large scientific research grant proposal.

Teaching
Tutoring and replacement lectures in various disciplines ranging from Quantum Optics to Thermodynamics.

Programming & Data Analysis

- **Python:** Advanced skills. Developed one open-source package (specialized physics library, see www.github.com/dlentrodt/) and contributed to two non-public packages.
- Basic skills in **Matlab, Java, C++, Bash, LaTeX, Excel**.
- **Data analysis:** Advanced skills in Bayesian Data Analysis.

Languages

German (mother tongue), English (C2; undergraduate studies at a British university), French & Spanish (basic skills)

Academic Achievements

Recent awards

Otto-Hahn-Medal 2023
National prize awarded by the Max Planck Society for outstanding PhD Thesis.

Finalist of the SAMOP Dissertation Prize 2024
Competition of 4 selected candidates for a national PhD Thesis prize awarded by the German Physical Society (DPG).

Publications

Own publications
9 Publications in top tier peer reviewed journals, including, e.g., Nature & Science. First author publications in Physical Review Letters, Physical Review X & Journal of Chemical Physics. Full list available at dlentrodt.github.io.

Peer review
Served as an expert reviewer for top tier journals including Nature, Physical Review Letters, Nanophotonics, and more.

Extracurricular activities

- Long history in various sports, e.g., served as treasurer for the Cambridge University Kickboxing Society and competed in Varsity matches vs. Oxford in 3 disciplines (first team in skiing & kickboxing, second team in judo).
- (Co-)Organisation of events, e.g., a winter school for Heidelberg PhD students, regular football matches between physics lecture participants and research groups in Freiburg, and more.